

■ MainSolve

`MainSolve[eqns]` is the underlying function for transforming systems of equations. `Solve` and `Eliminate` call it.

The equations must be of the form $lhs == rhs$. They can be combined using `&&` and `||`.

`MainSolve` returns `False` if no solutions to the equations exist, and `True` if all values of variables are solutions.

`MainSolve` rearranges the equations using certain directives.

`MainSolve[eqns, vars, elim, rest]` attempts to rearrange the equations `eqns` so as to solve for the variables `vars`, and eliminate the variables `elim`. The list `rest` can be included to specify the elimination order for any remaining variables.

`MainSolve` is essentially an internal routine. The functions `Solve` and `Eliminate` should be used for most practical purposes. ■ `MainSolve` works by creating a Gröbner basis from the equations. ■ The following options can be given:

<code>Mode -> Generic</code>	rational numbers and denominators depending on parameters can be introduced
<code>Mode -> Integer</code>	solutions must be over the integers; rational numbers cannot be introduced
<code>Mode -> Modular</code>	equality is required only modulo an integer
<code>Mode -> Rational</code>	rational numbers can be introduced (default)
<code>Method -> 1</code>	do one pass of elimination of variables only
<code>Method -> 2</code>	generate a disjunction of two equations to account for cases in which an expression can be zero
<code>Method -> 3</code>	solve the final equations for their roots

■ See also: `Solve`, `Eliminate`, `Reduce`, `SolveAlways`.